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ARTICLE

RAD-Seq Refines Previous Estimates of Genetic Structure in Lake Erie Walleye

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Abstract

Delineating population structure helps fishery managers to maintain a diverse "portfolio" of local spawning populations (stocks), as well as facilitate stock-specific management. In Lake Erie, commercial and recreational fisheries for Walleye *Sander vitreus* exploit numerous local spawning populations, which cannot be easily differentiated using traditional genetic data (e.g., microsatellites). Here, we used genomic information (12,264 polymorphic loci) generated using restriction site-associated DNA sequencing to investigate stock structure in Lake Erie Walleye. We found low genetic divergence (genetic differentiation index $F_{ST} = 0.0006–0.0019$) among the four Lake Erie western basin stocks examined, which resulted in low classification accuracies for individual samples (40–60%). However, more structure existed between western and eastern Lake Erie basin stocks ($F_{ST} = 0.0042–0.0064$), resulting in greater than 95%

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classification accuracy of samples to a lake basin. Thus, our success in using genomics to identify stock structure varied with spatial scale. Based on our results, we offer suggestions to improve the efficacy of this new genetic tool for refining stock structure and eventually determining relative stock contributions in Lake Erie Walleye and other Great Lakes populations.

Maintaining within-species variation creates a "portfolio" of population diversity that helps to buffer against fishing pressure and environmental fluctuation (Schindler et al. 2010; DuFour et al. 2015). However, preserving this diversity can be difficult in fish populations that are supported by discrete local spawning populations (i.e., stocks) but are harvested from a combined (mixed) fishery elsewhere. These types of mixed-stock fisheries can lead to the differential exploitation of multiple stocks, making management challenging. Successful management of mixed-stock fisheries crucially hinges on the ability to accurately identify the contribution of each stock to the fishery and the use of this information to ensure that all stocks are being managed sustainably (Vinther et al. 2004). Although management of many species has benefited from the determination of relative stock contributions to mixed-stock fisheries (e.g., Pacific salmonines: Shaklee et al. 1999; Dann et al. 2013; Atlantic Cod *Gadus morhua*: Ruzzante et al. 1999; Dahle et al. 2018), the same ability remains limited in other harvested fish populations, particularly the recreationally and commercially important fisheries of the Laurentian Great Lakes.

Most conspicuous in the Great Lakes basin has been the inability of fisheries biologists to consistently discriminate and identify the relative contributions of Lake Erie's local spawning populations of Walleye *Sander vitreus* (DuFour et al. 2015; Chen et al. 2017). Walleye is an economically important species that serves as the cornerstone to a large commercial fishery in Canadian waters of Lake Erie and an equally important recreational fishery in U.S. waters (Roseman et al. 2012; Walleye Task Group 2018). Being the most abundant large-bodied predator in the lake, Walleye has also been shown to play a major role in structuring Lake Erie's food web (Knight et al. 1984; Knight and Vondracek 1993; Kershner et al. 1999). Given this species' economic, ecological, and cultural importance, accurate stock assignment of Walleye to inform a mixed-stock management strategy has been a goal of Walleye research in Lake Erie for more than 40 years (Jarvis et al. 1978; Riley and Carline 1982).

Lake Erie's Walleye population is supported by multiple stocks that spawn in all three basins of the lake (Merker and Woodruff 1996; Gatt et al. 2004). However, the largest stocks appear to use habitat in the western basin to spawn, including the Sandusky River, Maumee River, Detroit River, and a shallow, open-lake reef complex (Figure 1; Wang et al. 2007; DuFour et al. 2015). Pre-recruit production among these western basin sites has been shown to vary annually (Roseman et al. 2005; DuFour et al. 2015). This variability may cause fluctuations in stock-specific abundances that could lead to differential exploitation rates among stocks (Frank and Brickman 2000). Therefore, the sustainability of the broader Lake Erie Walleye population is thought to depend on the maintenance of a diverse portfolio of local spawning populations (DuFour et al. 2015) and knowledge of the relative recruitment of each stock to the broader commercial and recreational fisheries (Chen et al. 2017; Faust et al. 2018).

Toward achieving these ends, numerous artificial and natural tagging approaches have been employed to match individual fish to specific stocks. Artificial tags (e.g., jaw, PIT, and acoustic) have been successfully used in Lake Erie to track movement and differentiate among individuals at the population level (Wang et al. 2007; Zhao et al. 2011; Vandergoot and Brenden 2014). Natural or "biomarker" tagging approaches (e.g., morphometrics, otolith microchemistry, and microsatellites), which rely on environmentally induced or inherited genetic markers, have been attempted during recent decades, with limited success, to identify natal origins of fish harvested in the open lake (Elsdon and Gillanders 2006; Elsdon et al. 2008). The use of morphometric and life history differences to determine natal origins among Walleye subpopulations was hindered by age-class effects (Watkinson and Gillis 2005). Likewise, the use of otolith microchemistry to determine natal origins was limited by interannual variation in water chemistry, and this approach has been only useful for differentiating individuals originating in the Sandusky River from other western basin spawning stocks (i.e., the Maumee River and Ohio reef complex could not be resolved; Chen 2016; Chen et al. 2017).

Genetic markers are a powerful tool for stock identification and have been used extensively to inform mixed-stock fisheries management (Begg et al. 1999a, 1999b; Utter and Ryman 2004). However, traditional genetic markers (e.g., microsatellites and mitochondrial DNA) have failed to yield consistent and conclusive stock definitions in Lake Erie's Walleye population (Stepien and Faber 1998; Strange and Stepien 2007; Stepien et al. 2012). For example, previous studies investigating Walleye genetic structure in Lake Erie using mitochondrial DNA and microsatellites have suggested that some stock structure exists both among and within basins (Stepien and Faber 1998; Strange and Stepien 2007; Stepien et al. 2009,

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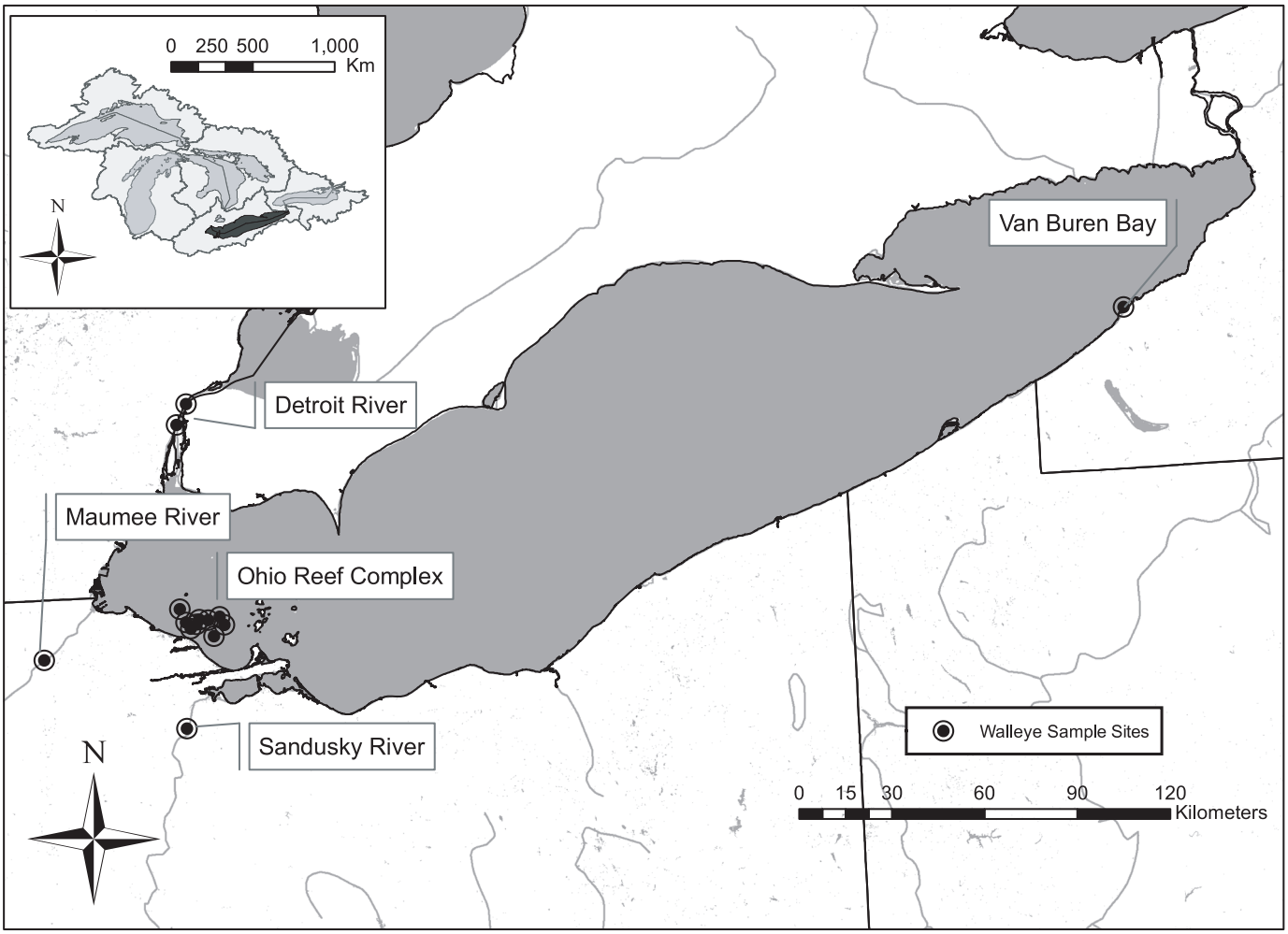

FIGURE 1. Map of the Great Lakes basin (upper-left inset), Lake Erie (dark gray), and locations of the Walleye local spawning populations sampled for this study. Black dots represent approximate sites of each sampled population. When more than one site was sampled for a given spawning population (e.g., the Ohio reef complex), multiple dots are circled to indicate that samples from these sites were pooled prior to analysis. (Map data source: U.S. Geological Survey 2010.)

2012). However, a blind experiment designed to test the ability of different genetic markers to assign 200 individuals of known origin back to their natal stock found that assignment success was low (generally < 50%), even when assigning fish at a basin scale (Johnson et al. 2004). Further assessment of two of western Lake Erie's largest spawning stocks (the Sandusky and Maumee rivers) and Van Buren Bay in the eastern basin showed low assignment rates of 56% and 22% for the Sandusky and Maumee River stocks, respectively, but a high assignment rate of 91% for Van Buren Bay (Stepien et al. 2012). In general, this lack of clarity surrounding genetic stocks of Walleye in Lake Erie has made it difficult for managers to construct strategies to ensure that all components of the portfolio of population diversity are sustainably managed.

Fortunately, recent advances in molecular techniques that facilitate efficient genotyping of thousands of markers hold the potential to clarify these ambiguous genetic relationships, as well as improve the accuracy of mixed-stock analysis. One such technique is restriction site-associated DNA sequencing (RAD-seq), which leverages restriction enzymes and high-throughput sequencing to discover and genotype thousands of single-nucleotide polymorphisms (SNPs) in any species (Andrews et al. 2016). Data from thousands of markers genotyped with RAD-seq have been successfully used to identify fine-scale structure in a number of species, including American lobster *Homarus americanus* (Benestan et al. 2015), Spangled Perch *Leiopotherapon unicolor* (Schmidt et al. 2018), and Chinook Salmon *Oncorhynchus tshawytscha* (Larson et al. 2014). Additionally, multiple studies have identified high-resolution markers from RAD-seq data that can be used to develop smaller high-throughput panels for stock identification (Larson et al. 2014; Hess et al. 2015). However,

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genomic techniques, such as RAD-seq, have rarely been utilized to inform fisheries management within the Great Lakes (but see Bernatchez et al. 2016).

Here, we applied RAD-seq to describe the genetic stock structure of Walleye in Lake Erie and to look for potential high-resolution variants that could be used for mixed-stock analysis. To test the utility of our RAD-seq approach, we focused mostly on western Lake Erie local spawning populations, as they have historically been thought to support Lake Erie's Walleye population (DuFour et al. 2015) and have been the most difficult to distinguish by using traditional molecular approaches (Johnson et al. 2004). However, we also contrasted these stocks with an eastern basin (out-group) stock to investigate levels of differentiation among versus within basins. Given that our study is among the first to use genomic techniques to inform management of fish populations in the Great Lakes, we discuss ways to increase the effectiveness of this tool for resolving stock structure and eventually for determining relative stock contributions in Lake Erie Walleye and other fish populations with limited spatial genetic variation.

METHODS

Samples.— State and federal agencies provided fin clips from Walleye collected during the spawning season at four spawning sites in western Lake Erie (Sandusky River, Maumee River, Detroit River, and the Ohio reef complex) and one site in the eastern basin (Van Buren Bay, New York; Figure 1; Table 1). We sought to analyze 100 fin clips from each site across 2 years for genetic analyses (i.e., ~50 fish per site per year). All fin clips were preserved in 95% ethanol in the field immediately after collection until processing for DNA extraction. All individuals were collected during the spawning season (April–May) at known spawning sites. Previous research has shown that Lake Erie Walleye tend to migrate to spawning sites only during the spring season to spawn, whereas they reside in open-water areas during the rest of the year (Bade et al. 2019). Given this movement behavior, the samples provided from all sites were presumed to represent active spawners. Fish from the Sandusky and Maumee rivers were captured via boat-mounted electrofishing by the Ohio Department of Natural Resources (ODNR)–Division of Wildlife (DOW) across the 2012 and 2013 spawning seasons. Fish from the Ohio reef complex were sampled across the 2012 and 2013 spawning seasons in conjunction with ODNR–DOW creel surveys conducted at boat ramps, which targeted anglers who had been fishing on the reef complex. U.S. Geological Survey Great Lakes Science Center personnel provided fin clips from Detroit River spawning individuals that were captured during the 2014 spawning season via electrofishing. We included this spawning population because, although it is not in Lake Erie, it is potentially an important contributor of Walleye to the Lake Erie fishery (DuFour et al. 2015). We also analyzed 25 fin clips from Walleye that were captured during the 2014 spawning season in Van Buren Bay, New York. These fish were captured by the New York State Department of Environmental Conservation using trap nets. Being geospatially separated from the other spawning sites, the fish from Van Buren Bay served as an out-group that might represent a different mitochondrial lineage than western basin stocks (Stepien and Faber 1998). All fish used in this study were handled either by state and federal agencies following department-specific animal-use guidelines.

Restriction site-associated DNA sequencing library preparation.— We extracted DNA with a Qiagen DNeasy Blood and Tissue Kit. We then processed extracts of DNA for library preparation and sequencing using a modified version of the RAD-seq protocol as described by Sovic et al. (2016). Briefly, each DNA extraction product was digested into fragments by two restriction enzymes: *Eco*RI and *Pst*I (i.e., digestion). The sticky ends of DNA fragments formed by the restriction enzymes were ligated with custom adaptors, which included a unique barcode

TABLE 1. Genetic diversity characteristics of adult Walleye collected at five Lake Erie spawning sites during 2012–2014 (H_O = observed heterozygosity; H_E = expected heterozygosity; N = number of Walleye fin clips analyzed). Effective population size (N_e) was estimated using linkage disequilibrium and corrected following the procedure of Waples et al. (2016). The N_e for all 345 western basin Walleye combined was 9,156.5, with a jackknifed 95% CI of 4,957.5–56,163.5.

Spawning site and year	N	H_O	H_E	N_e	Jackknifed 95% CI
Maumee River, 2012	45	0.119	0.118	4,943.3	1,299.8–∞
Maumee River, 2013	72	0.129	0.122	3,440.9	858.3–∞
Sandusky River, 2012	45	0.115	0.117	4,674.3	937.1–∞
Sandusky River, 2013	54	0.120	0.120	7,497.9	718.2–∞
Ohio reef complex, 2012	49	0.119	0.119	17,667.9	673.7–∞
Ohio reef complex, 2013	38	0.121	0.119	2,685.2	925.8–∞
Detroit River, 2014	42	0.116	0.118	7,111.5	398.8–∞
Van Buren Bay, 2014	25	0.114	0.116	3,181.7	215.9–∞

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for each individual. After the adaptor ligation, sample fragments ranging from 250 to 650 bp were extracted from agarose gels, subjected to quantitative PCR as a quality control measure, and PCR amplified to increase target sequence abundance before the high-throughput sequencing. Individual RAD-seq libraries were generated in 21 separate batches (2–30 samples per batch). Where possible, samples from each site were included within individual batches to control for potential batch effects. Samples within batches were pooled in equimolar amounts, and pools were run as partial lanes (comprising ~5–40% of a given lane, depending on the number of individuals included) on nine Illumina HiSeq 2500 runs at the Genomic Shared Resource at The Ohio State University (OSU) Comprehensive Cancer Center (CCC), Columbus. Paired-end (PE) run lengths ranged from PE75 to PE150 bp. This approach of mixing RAD-seq libraries with other library types on a lane helped to minimize any negative effects on sequencing quality arising from regions of low complexity in the RAD-seq libraries.

Bioinformatic analysis.— To identify SNP loci, we ran paired-end (R1 and R2 reads) raw sequence data separately through the AftrRAD bioinformatic analysis pipeline (Sovic et al. 2015) using default parameters, with the exception of changing the restriction site to be "TGCAG" to reflect the use of the *Pst*I enzyme. Prior to the analysis, reads exceeding 75 bp were trimmed to 75 bp to standardize read lengths for AftrRAD processing. The AftrRAD run removed any raw read that contained bases with low Phred quality scores (<20), had two or more incorrect bases at the restriction site, contained strings of the same base 20 or more bases long, contained the beginning of the P2 adaptor sequence, or had an average (nonzero) read depth less than 5. After loci were assembled, loci with three or more alleles that each had coverage greater than 10 reads within one or more individuals were considered paralogous and were removed. Genotypes were assigned at the remaining polymorphic loci (i.e., tags) for each individual that had at least five reads at the locus. Finally, the genotype data from the R1 and R2 reads were combined and we conducted preliminary population assignment analysis that allowed 0% (3,131 polymorphic loci), 5% (9,366 polymorphic loci), 10% (10,757 polymorphic loci), and 20% (12,264 polymorphic loci) missing data per locus across all samples (see Supplementary Materials available in the online version of this article). We found that 20% missing data provided the highest reassignment accuracy that was above the null assignment rate (Figure S1 available in the Supplement in the online version of this article) while still containing a level of missing data comparable to other studies (e.g., Paris et al. 2017); therefore, we chose to conduct all further analyses with this data set.

Population genetic analysis.— To describe the genetic architecture of the Lake Erie Walleye population, we estimated site-specific genetic diversity and genetic distance among the local spawning populations. We assessed whether individual loci conformed to Hardy–Weinberg equilibrium expectations at each locus within each stock by conducting chi-square tests in the R package pegas using the hw.test() function (Paradis 2010). Observed heterozygosity (H_O), expected heterozygosity, and inbreeding coefficient (F_{IS}) were estimated using the basicStats() function of the diveRsity package in R version 3.3.3 (Keenan et al. 2013; R Core Team 2015). Effective population size (N_e) at each sampled site was calculated using the linkage disequilibrium method in N_eEstimator (Do et al. 2014), with a minimum acceptable allele frequency of 0.02 (i.e., to avoid upward bias of N_e estimates) and all other parameters were set to default conditions. The N_e calculations using the linkage disequilibrium method can be biased slightly downward when individuals from multiple cohorts are included in the sample due to a slight Wahlund effect (7% downward bias on average; Waples et al. 2014). This small bias should not greatly affect the interpretation of the N_e results. However, because N_eEstimator assumes closed populations, which is not the case for western basin samples, N_e was also estimated for all western basin populations grouped together. We corrected estimates of N_e to account for physical linkage by chromosomes using the following equation outlined by Waples et al. (2016),

$$\frac{\hat{N}_e}{N_e} = [0.098 + 0.219 \cdot \log_e(24)],$$

where 24 is the number of haploid chromosomes in Walleye, N_e is the uncorrected estimate of effective population size, and $\hat{N}_e$ is the corrected estimate of effective population size. We assessed genetic differentiation among sites using a combination of standard *F*-statistics, analysis of molecular variance (AMOVA), maximum-likelihood ADMIXTURE analysis, and discriminant analysis of principal components (DAPC). Locus-specific genetic differentiation index F_{ST} and pairwise F_{ST} values with bias-corrected bootstrapped 95% CIs (number of bootstraps = 500) were calculated between all sample groups to infer the level of differentiation using the diffCalc() function in the diveRsity package (Weir and Cockerham 1984). Statistical significance of each pairwise value of F_{ST} was estimated using log-likelihood *G*-statistics in GenoDive version 2.0b27, and *P*-values were adjusted using Benjamini–Hochberg corrections (Benjamini and Hochberg 1995; Meirmans and Van Tienderen 2004). To estimate the percent of overall differentiation explained by basin, an AMOVA was conducted in Arlequin version 3.5, with basin (western versus eastern) as the grouping factor (Excoffier and Lischer 2010).

Genetic clustering without a priori assumptions of population was evaluated using ADMIXTURE (Alexander

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et al. 2009). ADMIXTURE was run on the complete data set of individuals and all bi-allelic loci, with default settings for each value of K (number of populations; $K = 1$–9). The optimal K-value was determined with the default cross-validation procedure built into ADMIXTURE, and results were visualized in ggplot2 in R (Wickham 2009). As a second evaluation of genetic clustering, we assessed variation among samples by using DAPC deployed through the Adegenet package in R (Jombart 2008; Jombart et al. 2010). To avoid overfitting the DAPC models, the optimal number of retained principal components was estimated using alpha scores, which are a measure of the difference between the observed proportion of successful reassignment in the present analysis and assignment to random groups. Estimates of stock membership for each individual were then estimated from the optimal DAPC and visualized in DAPC biplots.

Outlier analysis.— We conducted outlier analyses in BayeScan version 2.1, Arlequin version 3.5, and OutFLANK version 0.2 (Foll and Gaggiotti 2008; Excoffier and Lischer 2010; Whitlock and Lotterhos 2014) to identify loci potentially under selection that could be used as markers in a mixed-stock analysis. Three different methods were used because each test has unique biases and strengths, and therefore, comparing outliers among multiple methods can help to avoid false positives. BayeScan takes a Bayesian approach wherein evidence for selection is evaluated using Bayes factors and multiple comparisons are corrected using a false detection-rate calculation. Arlequin estimates outliers by calculating P-values for all loci under neutrality and the quantile of the observed F_{ST} in the overall distribution. We ran all outlier tests twice: once for all spawning sites and once with the eastern basin site (Van Buren Bay) removed (i.e., western basin populations only). BayeScan was run using a priori odds of 1,000, and all other parameters were held at default settings. Loci that were found to have a $\log_{10}$ q-score less than −2 and a false detection rate less than 0.01 were considered to be candidates for divergent selection among sites and putatively informative for population reassignment in a mixed-stock analysis. Outlier tests in Arlequin were run using a hierarchical island model, 100 simulated demes, and 20,000 coalescent simulations. Outliers were defined as candidates for divergent selection if they had an F_{ST} P-value of less than 0.01. Finally, outlier tests in OutFLANK were conducted using only bi-allelic loci and default settings.

Stock reassignment accuracy.— Incorporating "training" and "holdout" data sets when assessing assignment power is vital for ensuring that assignment results are not upwardly biased (Anderson 2010; Waples 2010; Chen et al. 2018). We took this training and holdout approach to evaluate whether individuals collected at the various spawning sites (see Figure 1) could be accurately reassigned to their site of capture. To do so, we used the R package assignPOP (Chen et al. 2018) to assign randomly chosen individuals from each local spawning site to training groups in one of three proportions (50, 70, and 90%), with the remaining individuals being used as test (i.e., holdout) individuals. All of the loci (with up to 20% missing data) from the randomly chosen training individuals were used to build a training model ($N = 3$ in total, using 50, 70, or 90% of individuals), which we subsequently tested by using all of the loci from independent holdout individuals.

These assignment tests were iterated 60 times for each proportion of training individuals. This procedure allowed us to evaluate variation in assignment accuracy and how different proportions of training individuals influenced the assignment results. Classification was conducted using a support-vector machine model, which has been shown to generate higher assignment accuracies than other models (Chen et al. 2018). We ran the reassignment test twice: once with every site and year run separately (i.e., Sandusky River, 2012; Sandusky River, 2013; Maumee River, 2012; Maumee River, 2013; Ohio reef complex, 2012; Ohio reef complex, 2013; Detroit River, 2014; and Van Buren Bay, 2014) and once with all western basin sites grouped together to test whether western and eastern basin stocks could be accurately delineated.

RESULTS

Sequencing and Filtering

In total, 1,955,670,361 reads were obtained; of these, 1,063,760,763 passed all quality filters and were assigned unambiguously to one of the samples based on the barcode. The median value of reads assigned to each individual sample after filtering was 2,493,240, with a range of 732,899–8,879,440. Overall, 12,264 polymorphic loci passed all quality filters and were used in our analyses. Of these loci, 10,293 contained only a single SNP and the remaining 1,971 loci contained more than one SNP. The average genotype rate across all 370 individuals was 96% (SD = 4%; median = 98%), and only a single individual had more than 20% missing data (i.e., 33%). As such, we chose to include all individuals in subsequent analyses. Summary statistics for each locus are available upon request or at https://github.com/WesLarson-Lab/Chen_et_al_2019_data.

Genetic Diversity and Structure

Average heterozygosity was similar across all local spawning sites ($H_O = 0.114$–0.129), and N_e was estimated to be in the thousands to tens of thousands of individuals for every site (Table 1). When all western basin populations were grouped, which likely represents a better

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assumption for population connectivity than individual site estimates, N_e was 9,157, with a jackknifed 95% CI of 4,958–56,164.

We documented low levels of stock structure throughout the lake (global $F_{ST} = 0.0018$). Western basin sites, which included the Sandusky River, Maumee River, Ohio reef complex, and Detroit River, had low but statistically significant pairwise F_{ST} values (0.0004–0.0019) when evaluated with log-likelihood tests. Pairwise differences between years within sites were generally lower than differences among sites but were still significant (Table 2). Although log-likelihood tests were statistically significant, the bootstrapped 95% CIs for all pairwise F_{ST} estimates within the western basin, as well as for the interannual pairwise comparisons, included zero.

Pairwise F_{ST} values between Van Buren Bay (eastern basin) and the western basin stocks were larger than pairwise F_{ST} among sites in the western basin (average pairwise $F_{ST} = 0.005$; Table 2). All pairwise F_{ST} estimates between Van Buren Bay and western basin stocks were statistically significant when evaluated with log-likelihood tests. However, only two of the seven pairwise F_{ST} values had CIs that did not include zero (Table 2).

The results from an AMOVA revealed that grouping of samples by basin (western versus eastern) explained only about 0.5% of the variation in allele frequencies (Table 3). Although low, this variation was larger than the variation found among local spawning populations (stocks) within each basin (~0.0%), further indicating significant but weak genetic structure between basins. However, diversity within local spawning populations was high, with the majority of genetic variation being found within stocks (>99.5%; Table 3).

Findings from DAPC and ADMIXTURE analyses supported our F_{ST} and AMOVA findings. Although DAPC revealed discernable structure between Van Buren Bay (eastern basin) and all western basin sites, it was unable to consistently differentiate among sites within the western basin (Figure 2). ADMIXTURE also identified some structure between the two basins across multiple values of K; however, the results were less clear with this analysis relative to the others. Specifically, no obvious best value of K was found in ADMIXTURE based on cross validation, indicating low stock structure in the data set (Figure S2).

Outlier Loci

Most loci showed low F_{ST} values (Figure 3); however, all outlier detection methods did identify putative outlier loci (Table 4). When all sites were considered together, BayeScan identified three outlier loci, OutFLANK identified 19 outlier loci, and Arlequin identified 172 outlier loci, with only a single locus being identified by all three programs (also see Figure S4). The average F_{ST} values of all outlier loci identified by BayeScan, OutFLANK, and Arlequin were 0.033, 0.150, and 0.167, respectively (Table 4). When only western basin sites were evaluated, BayeScan and OutFLANK each identified a single (but different) outlier locus, whereas Arlequin identified 51 outlier loci (including the same outlier detected by BayeScan but not the locus identified with OutFLANK). The average F_{ST} values of all outlier loci identified for western basin sites were 0.027, 0.060, and 0.036 for BayeScan, OutFLANK, and Arlequin, respectively.

Stock Reassignment Accuracy

Reassignment accuracies varied among stocks and between years, as well as based on the spatial scale of the analysis (Figure 4). Correct overall reassignment using assignPOP and all available individuals was around 50% when 50, 70, and 90% of individuals were used in the training set (Figure 4A). However, in many cases, the assignment accuracies tended to improve as more individuals were added to the training set (e.g., Sandusky River, 2013; Ohio reef complex, 2013; Detroit River, 2014; and Van Buren Bay, 2014; Figure 4A). For the western basin stocks, correct reassignment was less than 50% for all stocks except the Sandusky River stock during 2013, which had a median correct assignment of 75%, which was consistent across training data sets. Van Buren Bay (eastern basin stock) had a higher reassignment accuracy than any of the western basin stocks, with a median reassignment accuracy that exceeded 80% when more than 70% of individuals were used in the training set (Figure 4A, rightmost panel). When all of the individuals from the four western basin stocks were grouped together (as a single western basin stock), overall reassignment accuracy (to a basin) exceeded 95% for both basins, although some western basin individuals were misclassified to the eastern basin population (Figure 4B).

DISCUSSION

Multiple studies have used traditional sources of genetic data (e.g., mitochondrial DNA, microsatellites, etc.) to document genetic structure in Lake Erie Walleye, but the conclusions drawn from those studies have been inconsistent among and within laboratories and variable among methods. Given these inconsistencies, we sought to determine whether a newer genomic technique, RAD-seq, could clarify stock structure, quantify genetic diversity, and facilitate accurate mixed-stock analysis of Lake Erie Walleye. Our findings revealed that genetic diversity at all sites was similar and that N_e was large (thousands to tens of thousands of individuals). Outlier tests revealed a small number of significant outlier loci between eastern and western basin stocks and, to a lesser extent, among sites within the western basin. Although we were able to

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TABLE 2. Genetic differentiation index F_{ST} values of adult Walleye collected at five Lake Erie spawning sites during the spawning season, 2012–2014. Four western basin sites (Maumee River [MR], Sandusky River [SR], Ohio reef complex [Reefs], and Detroit River [DR]) and one eastern basin site (Van Buren Bay [VB]) were sampled during 2012, 2013, and/or 2014. Pairwise values of Weir and Cockerham's (1984) theta (i.e., F_{ST}; below diagonal) and bootstrapped 95% CIs (above diagonal) are reported. Pairwise values with Benjamini–Hochberg adjusted *P*-values less than 0.05 are denoted with an asterisk; values with 95% CIs that do not include zero are in bold.

Site and year	MR 2012	MR 2013	SR 2012	SR 2013	Reefs 2012	Reefs 2013	DR 2014	VB 2014
MR 2012		−0.0008 to 0.0041	−0.0023 to 0.0042	−0.0016 to 0.0043	−0.0021 to 0.0036	−0.0022 to 0.0054	−0.0026 to 0.0044	−0.0006 to 0.0117
MR 2013	0.0013*		−0.0004 to 0.0047	−0.0008 to 0.0038	−0.0004 to 0.0044	−0.0013 to 0.0052	−0.0015 to 0.0043	**0.0012–0.0127**
SR 2012	0.0008*	0.0019*		−0.0024 to 0.0038	−0.002 to 0.0047	−0.002 to 0.0066	−0.0023 to 0.0051	−0.0003 to 0.013
SR 2013	0.0009*	0.0012*	0.0004*		−0.0017 to 0.0041	−0.002 to 0.0055	−0.0026 to 0.004	**0.0003–0.0120**
Reefs 2012	0.0004*	0.0017*	0.0009*	0.0009*		−0.0027 to 0.0054	−0.0028 to 0.0045	−0.0004 to 0.0116
Reefs 2013	0.0012*	0.0017*	0.0017*	0.0014*	0.0008*		−0.002 to 0.0063	−0.0003 to 0.0129
DR 2014	0.0006*	0.0013*	0.0010*	0.0006*	0.0006*	0.0015*		−0.0013 to 0.0115
VB 2014	0.0049*	**0.0064***	0.0059*	**0.0055***	0.0052*	0.0056*	0.0042*	

successfully differentiate between spawning sites in western versus eastern Lake Erie, we found no genetic structure among the four local spawning populations (stocks) sampled in the western basin. Overall, our study showed promise for RAD-seq as a means to discern between Walleye stocks at the basin scale in Lake Erie, which could be used to better understand and manage the commercial and recreational fisheries that this population supports. Below, we discuss our findings in relation to previous genetic studies, as well as offer suggestions that could improve the use of modern genomic approaches to discern stock structure in populations that have low genetic structure (e.g., western basin spawning populations).

Genetic Structure in the Western Basin

Genetic structure among the four spawning populations (Maumee, Sandusky, and Detroit rivers and the Ohio reef complex) in the western basin of Lake Erie was low during both study years. Although statistically significant positive F_{ST} values were identified among all sites, the power to detect differences with our current methods was high. Therefore, we question the biological significance of these results, especially considering that inconsistencies were found in the two methods that we used to assess significance. Specifically, our log-likelihood approach found slight but observable differences among western basin stocks, whereas our bootstrapped 95% CIs indicated that these differences were not always detectable. Because the latter bootstrapping approach provides a better indication of the replicability of this result (Cumming 2008; Lazzeroni et al. 2014), we give more credence to the notion that biologically meaningful genetic structure is lacking among western Lake Erie's spawning populations.

Our findings help to clarify prior genetic research conducted on Lake Erie Walleye. For example, our results support a previous finding by Strange and Stepien (2007), who used microsatellite data to conclude that western basin Walleye consisted of a single stock. In that study, no significant differences were found among western basin spawning sites, although the study had relatively low sample sizes and limited temporal replication. Recognizing these limitations, Stepien et al. (2012) expanded the Strange and Stepien (2007) data set by adding samples from five additional years. Using this larger data set, Stepien et al. (2012) concluded that the Sandusky and Maumee rivers represented two distinct genetic stocks. However, significant genetic differences between these two stocks were only found when all years were pooled. Additionally, reassignment accuracies in the Stepien et al. (2012) study were low for both the Maumee and Sandusky rivers (22% and 56%, respectively). These values are quite similar to our own, which were determined using thousands of loci (versus nine microsatellites) screened on large sample sizes of adult Walleye collected across the spawning season. Our results, therefore, contrast those of Stepien et al. (2012) and suggest that little, if any, genetic structure exists between the Maumee and Sandusky rivers. Although it is possible that a more thorough genome scan

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TABLE 3. Analysis of molecular variance results for adult Walleye collected at five spawning sites during the spawning season in Lake Erie, 2012–2014. Local spawning populations (stocks) were grouped by basin (western versus eastern), with the Maumee, Sandusky, and Detroit rivers and the Ohio reef complex comprising the western basin group and Van Buren Bay representing the eastern basin group. Bold values indicate significance ($P < 0.05$).

Source of variation	df	Percentage of variation
Between groups (western versus eastern basin)	1	0.49
Among stocks within groups	6	−0.04
Within stocks	732	**99.55**

would reveal loci under divergent selection that can differentiate these populations (Lowry et al. 2017), we were unable to find a large number of such loci in our study.

Genetic Structure between Eastern and Western Basins

Our highest estimates of genetic distance were between western basin spawning sites and Van Buren Bay in the eastern basin, which suggests the existence of at least two genetically different stocks in Lake Erie. This two-stock hypothesis was supported by our DAPC and assignPOP results, which showed consistent, accurate reassignment at the basin scale. However, ADMIXTURE did not clearly delineate between the eastern and western basin spawning populations and AMOVA results suggested that only about 0.5% of the variation in our data was explained by lake basin, indicating that the genetic difference between basins is still small. Earlier studies of Walleye genetic structure were unable to differentiate between western basin sites and Van Buren Bay (Strange and Stepien 2007; Stepien et al. 2009), whereas our findings are consistent with a recent study that also found significant genetic distance between Van Buren Bay and the Maumee and Sandusky rivers during all years and that could correctly reassign 91% of individuals from Van Buren Bay (Stepien et al. 2012). Collectively, these findings indicate that genomic approaches, such as RAD-seq, could be useful for identifying genetic differences between western and eastern basin Walleye populations. Once these differences are quantified, other, higher-throughput molecular resources (e.g., genotyping-in-thousands by sequencing [GT-seq] and RAD capture [Rapture] panels) could be developed to determine the relative contributions of western versus eastern basin stocks to Lake Erie's mixed-stock fisheries (Campbell et al. 2015; Ali et al. 2016; Meek and Larson 2019).

Mechanisms for Low Population Differentiation

Two mechanisms can help to explain our inability to discern western basin spawning populations and the low

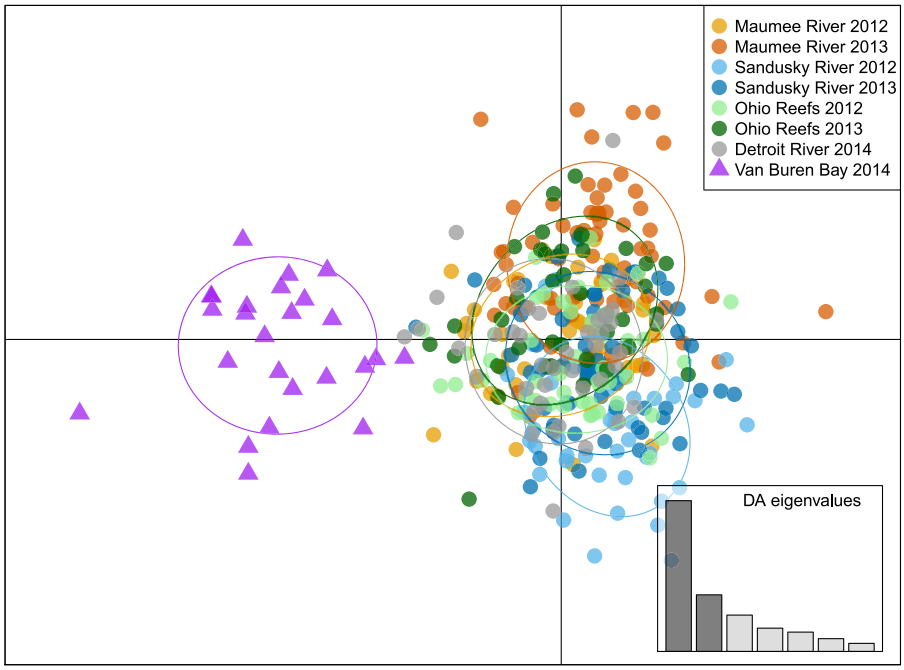

FIGURE 2. Individual-based discriminant analysis of principal components (DAPC) for adult Walleye collected at four western Lake Erie spawning sites (Maumee, Sandusky, and Detroit rivers and the Ohio reef complex) and one eastern Lake Erie spawning site (Van Buren Bay) during 2012–2014. The DAPC was based on the first 50 principal components (PCs), with clusters defined as each sampled spawning site; the first two DAPC axes are portrayed. Inertia ellipses represent a graphical summary of coordinate variance used to define each group and should not be interpreted as CIs. The relative magnitude of the discriminant analysis (DA) eigenvalues for the first seven PC axes is presented in the inset box.

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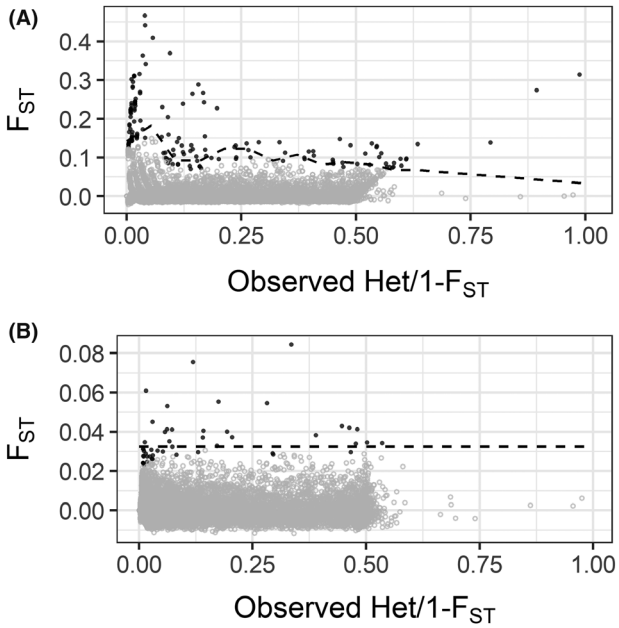

FIGURE 3. Genetic differentiation index F_{ST} outlier loci detected in adult Walleye collected at four western Lake Erie spawning sites (Maumee, Sandusky, and Detroit rivers and the Ohio reef complex) and one eastern Lake Erie spawning site (Van Buren Bay) during 2012–2014. Analyses were conducted in Arlequin assuming a hierarchical model of migration for **(A)** all populations and **(B)** only western basin populations. Significant F_{ST} values ($\alpha = 0.01$) are highlighted as filled black circles; F_{ST} is a locus-specific measure of genetic divergence among populations, and "Observed Heterozygosity (Het)/1 – F_{ST}" is a measure of heterozygosity per locus. The upper 99% quantile is plotted as a dashed line.

TABLE 4. Results of outlier analyses conducted using loci from adult Walleye collected at five spawning sites during the spawning season in Lake Erie, 2012–2014. The western basin sites consisted of the Maumee, Sandusky, and Detroit rivers and the Ohio reef complex. One additional site was sampled in the eastern basin (Van Buren Bay). The number of genetic differentiation index F_{ST} outliers across all five sites, as well as only the four sites within the western basin, was identified using variance in allele frequencies from a simulated null distribution of F_{ST} values under an island model (BayeScan), hierarchical island model (Arlequin), and inferred distribution of null F_{ST} (OutFLANK).

Outlier detection method	Number of outliers	Average F_{ST} of outliers
BayeScan, all sites	3	0.033
BayeScan, western basin only	1	0.027
OutFLANK, all sites	19	0.142
OutFLANK, western basin only	1	0.061
Arlequin, all sites	172	0.167
Arlequin, western basin only	51	0.036

level of differentiation that exists between the eastern and western basins. First, while nonmolecular approaches have shown clear evidence of natal homing and spawning site fidelity among these populations, low levels of stock mixing during the spawning season have also been documented (e.g., Zhao et al. 2011; Faust et al. 2018; Hayden et al. 2018). For example, using otolith microchemistry, Chen (2016) estimated that 1–3% of spawners in the Sandusky River had natal origins elsewhere. Likewise, using an acoustic telemetry (artificial tagging) approach, Hayden et al. (2018) estimated spawning site fidelity of Maumee River Walleye to be only about 70%. The reported percentages of straying individuals in these western basin populations are small, but they likely translate to a large absolute number of strays, as the western Lake Erie Walleye spawning population is estimated at 30–40 million individuals and was projected to be 45 million by 2019 (DuFour et al. 2015; Walleye Task Group 2019). Therefore, it is conceivable that sufficient straying during the spawning season is occurring to minimize genetic differentiation among the groups while at the same time having negligible impacts on ecological inferences regarding homing behavior or spawning site fidelity (Waples 1998).

Second, spawning stocks could have limited genetic differentiation at neutral loci but still differentiate at loci under natural selection. Because the population size of Walleye in Lake Erie is so large, natural selection may be a stronger driver of genetic differentiation than drift (Hedrick 2005a), especially if there are differential selective pressures among spawning sites. Habitat conditions do vary substantially among our spawning sites, both within and between basins. Therefore, natural selection pressures specific to each spawning site may be strong enough to cause populations to diverge. For example, large spring precipitation events have been suggested to negatively affect larval Walleye survival in western basin streams but enhance survival of larvae on the Ohio reef complex (Mion et al. 2009; DuFour et al. 2015). Additionally, because Walleye generally remain within their basin of origin for the first 1–3 years of life, young Walleye from the eastern and western basins should experience substantially different habitat conditions (e.g., temperature, light, and prey availability) during the period when they are most vulnerable to natural mortality (Sifa and Mathias 1987; Wang et al. 2007).

Even with thousands of loci, genetic evidence of natural selection can be difficult to document. This is because the vast majority of loci are not under selection and hence have low genetic differentiation among all populations in the absence of strong genetic drift. Therefore, loci under selection can be easily missed with reduced-representation methods, such as RAD-seq (Lowry et al. 2017). Although we were able to identify some F_{ST} outliers, our analysis did not provide strong support for adaptive divergence

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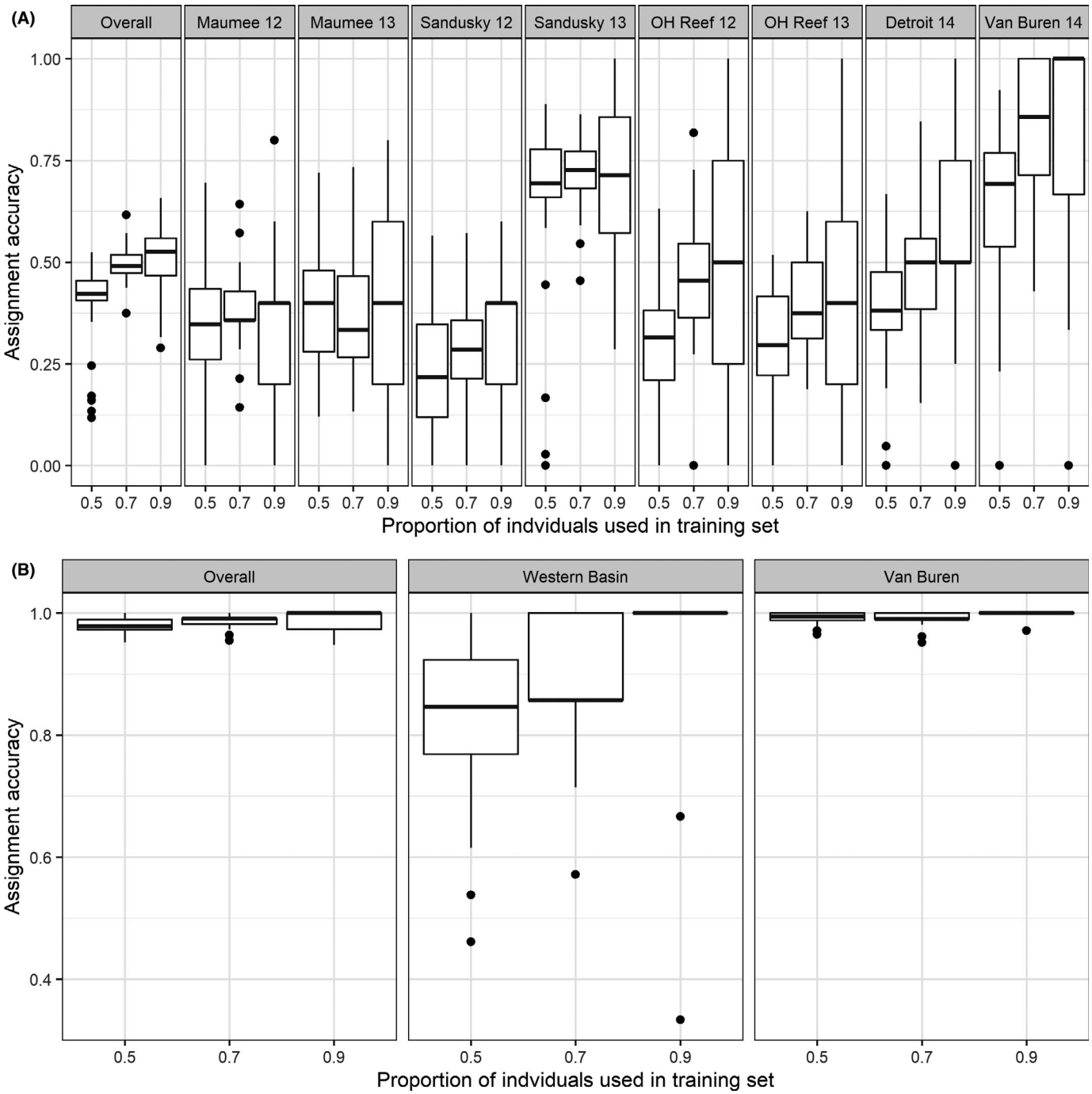

FIGURE 4. Reassignment accuracies of adult Walleye collected at four western Lake Erie spawning sites (Maumee, Sandusky, and Detroit rivers and the Ohio reef complex) and one eastern Lake Erie spawning site (Van Buren Bay) during 2012–2014: **(A)** reassignment accuracies obtained when each site × year combination was treated as a separate group, with the overall average reported in the left-most panel; and **(B)** reassignment accuracies obtained when local spawning populations were grouped by basin (western versus eastern), with the average presented in the left-most panel. Reassignment accuracy was determined using a support-vector machine algorithm, with training samples for each grouping consisting of 50, 70, or 90% of the collected individuals (chosen randomly). The remainder of individuals (50, 30, or 10%) was used as the test (holdout) data set to determine reassignment accuracy. Box plots portray medians (thick black line), interquartile ranges (ends of boxes), and outliers (black dots).

among sites in the western basin because F_{ST} values at these loci were still relatively low (~0.01–0.06) and false positives are common for outlier detection analyses unless selection is strong (Narum and Hess 2011). Furthermore, owing to the small number of loci identified, we were unable to properly cross validate the assignment accuracy

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of a data set including only outlier SNPs (Waples 2010). Thus, while the outliers that we detected differentiated the populations in the current data set to some degree, it is not possible to state conclusively whether they are consistently diagnostic of population structure or simply false positives that have drifted slightly more than other markers. Alternative molecular approaches that sample larger portions of the genome, such as genome resequencing, may do a better job of identifying markers under selection that could be used to distinguish between Walleye populations (Fuentes-Pardo and Ruzzante 2017).

Improving Genomic Approaches for Great Lakes Fishery Management

Despite the successful use of genomic approaches to the benefit of fisheries management in marine ecosystems (e.g., Ward 2000; Schwartz et al. 2007), we are only just beginning to understand the value of this tool for application to Great Lakes populations, such as Lake Erie Walleye. Below, we discuss several study limitations and identify ways in which tools such as RAD-seq and population reassignment methods could be improved to help agencies better understand stock structure and estimate relative contributions of genetic stocks to their valued fisheries.

A key limitation to our study was the lack of comparison among spawning stocks in the western basin of Lake Erie and spawning stocks in other Lake Erie basins. Most of the major western basin spawning stocks were sampled, whereas only one (Van Buren Bay) of at least six known major eastern basin stocks was investigated. Similarly, none of the central basin stocks (e.g., Grand and Cuyahoga rivers, Ohio) was included in this analysis. Given that our findings both supported (eastern versus western basin discrimination) and refuted (western basin discrimination) the findings of Stepien et al. (2012), we cannot speculate as to how estimates of genetic structure in the additional spawning populations will differ from those in past studies until the analyses are conducted. Therefore, to truly develop a high-quality, systemwide stock assignment program, we strongly encourage studies that expand the spatial scale of our investigation, both within and among basins. Additionally, we found that genetic structure in Lake Erie Walleye is shallow. Thus, the number of individuals used in stock reassignment training data sets needs to be high (Chen et al. 2018) and any population variance identified should be tested on an independent data set to avoid overfitting and biased results (Waples 2010).

Our study is one of the first in the Great Lakes to use genomic techniques to assess stock structure (also see Bernatchez et al. 2016 and Elias et al. 2018) and indicates that high-throughput sequencing methods, such as RAD-seq, can improve resolution in systems with shallow genetic structure and evaluate genetic diversity. For example, our analysis suggests that the N_e/N ratio of Walleye is similar to those of many marine fishes (Hedrick 2005b) but was not robust enough to confirm this finding without further research to assess the accuracy of both the N_e and N estimates. However, we emphasize that additional techniques and research exist, which could improve upon what we have shown here. For example, higher genomic coverage would increase the probability of detecting informative loci. Higher genomic coverage could be achieved by using alternate enzymes during RAD-seq preparation that increase the number of cut sites or by conducting full-genome resequencing (Fuentes-Pardo and Ruzzante 2017; McKinney et al. 2017). Once the most informative loci for stock delineation are identified, they can be used to construct resources, such as GT-seq and Rapture panels, which can quickly and consistently genotype thousands of individuals and in turn greatly increase sample sizes and statistical confidence (Meek and Larson 2019). These panels could then be used independently (if the resolution is high enough) or in conjunction with other techniques, such as otolith microchemistry, to successfully delineate populations (Chen et al. 2017). Therefore, further assessment of Walleye with higher-coverage RAD-seq or full-genome resequencing could be highly valuable for developing tools to conduct mixed-stock analysis of Walleye in the Great Lakes.

Conclusions

Although genetic structure appears to exist within the Lake Erie Walleye population, the ability to use this structure to discriminate among spawning stocks is spatial scale dependent. Even with thousands of polymorphic loci, stock structure within the western basin was difficult to identify, whereas discrimination between lake basins (western versus eastern) was possible. Despite our inability to consistently discriminate the western basin populations genetically, it is important to note that these stocks have biologically meaningful differences in life history (e.g., natal homing behavior; Chen 2016) and annual recruitment (DuFour et al. 2015). Therefore, delineation may be possible using a combination of environmental markers (e.g., parasites and otolith microchemistry) and molecular markers (MacKenzie and Abaunza 2005; Chen et al. 2017, 2018).

Genetic and genomic tools have revolutionized fisheries management and have made it possible to monitor fish stocks with a higher degree of precision than ever before possible (Dann et al. 2013). Our results demonstrate the ability to use modern genomic techniques, such as RAD-seq, to evaluate patterns of differentiation in populations with shallow genetic structure and highlight the importance of re-evaluating stock structure as technology advances. The inconsistency of past genetic studies has made it difficult for managers to construct strategies that ensure the sustainability of Lake Erie Walleye. Our study

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helps to resolve this issue by identifying a low degree of genetic differentiation with a high level of confidence. Studies like ours, which re-evaluate stock structure by using more powerful techniques, can play an important role in adaptive management by providing new, updated information about stock delineation. Genomics plays an important role in informing fisheries management, and our ability to delineate populations will only improve as molecular resources increase and as methods are refined for inland fisheries.

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